

Master's Degree in Computational Social Sciences
2025-2026

Master Thesis

The Ebb and Flow of Cooperation: Modeling the Genetic Spread of Altruist Traits Through Multilevel Selection

Ignacio Jiménez Ferrer-Sama

Tutor Anxo Sánchez
Madrid, 19 de junio de 2026

AVOID PLAGIARISM

The University uses the **Turnitin Feedback Studio** program within the Aula Global for the delivery of student work. This program compares the originality of the work delivered by each student with millions of electronic resources and detects those parts of the text that are copied and pasted. Plagiarizing in a TFM is considered a **Serious Misconduct**, and may result in permanent expulsion from the University.



[Include this code in case you want your Master Thesis published in Open Access University Repository]

This work is licensed under Creative Commons **Attribution – Non Commercial – Non Derivatives**

SUMMARY

Traditional natural selection struggles to explain the evolution of altruistic behavior, defined as "individual cost and group benefit". This paradox is robustly addressed by Multilevel Selection theory (MLS), demonstrating how group-level benefits can overcome individual free-riders. Despite its theoretical power, MLS suffers from a lack of mathematical models capable of precisely defining the structural conditions required for group-level selection to dominate. The present work addresses this gap by utilizing an agent-based computational model that simulates the evolutionary dynamics of cooperation across overlapping sets. Through Random Forest regression and Locally Estimated Scatterplot Smoothing (LOESS), the analysis evaluates the impact of network density, cost-benefit ratios and set mutation rates on the success of altruistic traits. The synergy metric is introduced to represent altruistic actions that create more total benefit when the number of recipients increases. The results show that both the cost-benefit ratio and set mutation have a strong, linear effect on cooperation success: the cost-benefit ratio increases success, while set mutation decreases it. Synergy has shown to provide high resilience for cooperators, as well as having a great augmenting effect on the cost-benefit ratio. Finally, sparse network density appears to mitigate the negative effects of set mutation, while synergistic environments require a minimal network density for cooperation to succeed.

Keywords: Multilevel Selection, Altruism, Agent-Based Modeling, Evolutionary Game Theory, Social Network Analysis, Behavioral Evolution

CONTENTS

1. INTRODUCTION.	1
2. THEORETICAL FRAMEWORK	3
2.1. The Original Explanation: ‘Naive’ Group Selection	3
2.2. Resistance to Multilevel Selection	4
2.3. Mathematical models	5
3. THE MODEL	7
3.1. Model Definition	7
3.2. Methodology	9
3.3. Analysis.	11
3.4. Variable Importance	11
3.4.1. Decision Tree Analysis	12
3.4.2. 2D LOESS Plots	16
3.4.3. b/c Ratio vs K/M Ratio	17
BIBLIOGRAPHY.	20

LIST OF FIGURES

3.1	<i>Random Forest Variable Importance</i>	11
3.2	<i>Decision Tree. Synergy = 0</i>	13
3.3	<i>Decision Tree. Synergy = 0.5</i>	14
3.4	<i>Decision Tree. Synergy = 1</i>	15
3.5	<i>b/c Ratio vs Synergy</i>	16
3.6	<i>b/c Ratio vs K/M Ratio. Synergy = 0</i>	17
3.7	<i>b/c Ratio vs K/M Ratio. Synergy = 0.5</i>	17
3.8	<i>b/c Ratio vs K/M Ratio. Synergy = 1</i>	18

1. INTRODUCTION

Traditional natural selection struggles to explain the evolution of altruist behavior. Because traits that benefit all members in a group don't provide a relative fitness advantage to the individual possessing them, their frequency can't increase within said group. Since most behaviours impose a cost on the individual, "free-riders" that receive the benefit without paying the cost are meant to out-reproduce altruist individuals (van Wijngaarden, 2025).

However, altruistic traits continually evolve across all scales of life. Examples can be found from virus strains reducing their virulence to avoid killing their hosts and ensure transmission (Marín et al., 2026), or the microalga *Prymnesium parvum* secreting defensive toxins as an exploitable public good (Marín et al., 2026), to the honey bees' highly cooperative societies (Wilson, 1997), and, as we will argue, even the massive organizational structures of human societies (van Wijngaarden, 2025; Wilson et al., 2023).

This contradiction between evolutionary theory and observed reality has been a subject of intense observational, experimental and theoretical work across a variety of scientific disciplines (Gavrilets, 2015). As researchers have worked to solve the puzzle, multiple explanatory frameworks have been proposed, resulting in one of the most contentious debates in the history of the field (Greene et al., 2025).

Multilevel selection (MLS) offers a useful framework for thinking about the problem of altruism because it allows robust evolutionary logic to explain phenomena beyond individual survival and fitness.

Despite having strong empirical support and a mathematically sound framework, Multilevel Selection remains largely ignored or misrepresented in undergraduate biology textbooks. An academic review (Greene et al., 2025) found that six out of nine widely used textbooks define natural selection strictly as a process that occurs at the level of the individual. In these books, mentions of MLS were limited to a small subsection, treated as a niche theory for solving animal behavioral paradoxes, and failing to reflect the vast array of real-world evolutionary phenomena that the theory helps explain.

Mathematical models of multilevel selection play a crucial role in the formalization of MLS by offering a rigorous framework to test theoretical foundations and interpret empirical results from complex systems (van Wijngaarden, 2025). However, a persistent challenge in the field is a notable lack of models capable of clearly revealing the balance between these antagonistic forces of selection or precisely defining the mathematical conditions required for the dominance of selection at the group level (van Wijngaarden, 2025). In multilevel selection theory, this balance has often only been studied superficially due to the lack of widely applicable stochastic models. While conditions for the dominance of group-level selection have been observed in empirical systems, the broader

applicability of these specific conditions remains limited (van Wijngaarden, 2025). This thesis aims to address the gap by mapping the effect of certain structural and environmental parameters on the development of cooperation, not only with the goal of helping foster cooperative behavior within human institutions, but also to gain insight on the nature of altruism and selfishness as adaptive strategies.

2. THEORETICAL FRAMEWORK

2.1. The Original Explanation: ‘Naive’ Group Selection

Thinking of higher-level units such as social groups in functional terms is an ancient intellectual tradition that exists in all branches of science and in everyday thought (Wilson, 1998). In biology, this idea can be traced back to Darwin, and it was widely and uncritically accepted for decades. Biologists routinely assumed that animals would sacrifice their own reproductive success “for the good of the group” or even “the species”, the most famous champion of this theory being V. C. Wynne-Edwards (1962).

These early formulations of group selection were mathematically naive and lacked a rigorous mechanism to explain how a group benefit could outplay the local advantage of the selfish free-rider. Eventually, the theory was so thoroughly mocked and rejected that it became a textbook example of flawed evolutionary thinking (Borrello, 2005).

In the wake of this rejection, biologists developed other frameworks to solve the paradox of altruism. Here we will cover three: kin selection, selfish gene theory and evolutionary game theory. These alternatives have proven their immense worth by correctly predicting how social behaviors evolve without needing to invoke “the good of the group” (van Wijngaarden, 2025; Wilson, 1998). However, as I will defend, these theories are not mutually exclusive and can be regarded as different perspectives over the same process (Wilson, 1997).

Kin selection was originally developed by Hamilton (1964), this theory proposes that altruism can evolve if the cost to the actor is lower than the benefit created multiplied by their genetic relatedness of the recipient ($rb > c$). In this view, self-sacrifice is just a genetic strategy, initially predicting that altruism should be strictly confined to genealogical relatives (Wilson, 1998).

Built upon Hamilton’s theory, the selfish gene theory (Dawkins, 1976) is a broader conceptual framework which posits that the gene, and not the individual, is the fundamental unit of selection. Individuals are viewed merely as temporary vehicles built by genes to replicate themselves (Wilson, 1997). All behaviors that evolve, no matter how apparently altruistic, ultimately serve only this sole purpose. Kin selection fits this view, as it could be defined as “helping your own genes in other people’s bodies”. In this new light, Hamilton’s ‘ r ’ coefficient changed from a measure of relationship via descent to a broader statistical measure of genetic similarity: the covariance between the genotype of each potential actor and the average genotype of their recipients (Fletcher & Zwick, 2007). This statistical association was found to be a crucial factor in the evolution of altruism, in other words, cooperators need to cluster together in order to succeed. Critically, this clustering can be reached through many mechanisms, even those involving random

grouping.

Finally, evolutionary game theory models how different behaviors or strategies interact in a population. This framework designs models with games like the N-player Prisoner's Dilemma to map out the conditions under which cooperative strategies are evolutionarily stable. Papers in this field consistently mention different forms of cooperator clustering, assortment or "spatial structure" as crucial to the game's outcome (Fletcher & Zwick, 2007; Szolnoki & Perc, 2009; Tarnita et al., 2009; Tekwa et al., 2015).

Relegated to the margins for decades due to the stigma of naive group selection, a fourth framework experienced a resurgence in the 1990s: multilevel selection. MLS revived the concept of group-level adaptation, but this time it was armed with a mathematical mechanism to explain how altruism evolves in metapopulations (populations subdivided into multiple groups) (Wilson, 1997). Ultimately, the theory demonstrated that the only requirement for an altruistic trait to evolve is a sufficiently high degree of between-group variation around said trait (van Wijngaarden, 2025; Wilson, 1998). Again, this idea is exactly equivalent to the concepts of 'clustering' or 'assortment' mentioned above.

Wilson (1997) summarizes the mechanics of MLS brilliantly:

"A mutant gene appears that causes the individual to behave in a way that benefits everyone in the group, including itself. There are no fitness differences within the group containing the mutant gene, but the entire group contributes more progeny to the global population than do other groups, which causes the gene to increase in frequency. Every generation, groups will randomly vary in the frequency of the beneficial gene. Groups with the highest frequency will contribute the most progeny to the global population and therefore to the formation of new groups. Ultimately, the beneficial gene will evolve to fixation."

Or, in short: "Selfishness beats altruism within groups. Altruistic groups beat selfish groups. Everything else is commentary." (Wilson & Wilson, 2007).

2.2. Resistance to Multilevel Selection

Historically, critics dismissed MLS by claiming that group-level selection required extreme levels of genetic variation, only found in populations sharing genes like eusocial insects or those undergoing intense inbreeding (Wilson, 1997). However, modern multilevel selection theory demonstrates that strict genetic isolation is not a prerequisite for group selection. Two major factors refute this critique:

First, spatial assortment extends far beyond distinct, rigidly bound physical groups. What truly matters for selection is how individuals assort and interact with respect to a specific behavior or trait. This dynamic is captured by the concept of "trait groups", allowing us to define the ' r ' coefficient as between-group variance divided by the total variance *in the cooperative trait* (Fletcher & Zwick, 2007). This makes the level of

assortment (r) in a population be significantly higher than what would be predicted by genetic relatedness alone (Fletcher & Zwick, 2007).

Second, the evolution of social control mechanisms, such as moral norms, rewards, and punishments, force behavioral conformity within groups, which in turn increases phenotypic variance between groups. The phenotype is the actual expressed behavior or physical trait in an organism, as opposed to underlying genetics. Because evolutionary success is dictated by actual behaviors, phenotypic variation is the ultimate ingredient of natural selection, on the condition that it correlates with genotypic variation (Wilson, 1998). By using social control to erase the behavioral advantages of free-riders locally and amplify behavioral differences globally, populations can evolve highly sophisticated group-level adaptations without requiring extreme genetic variance or inbreeding (Wilson et al., 2023).

Social norms are often invoked as alternative, entirely independent explanations for altruist behavior (Wilson, 1998), but they cannot be fully explained without some form of group-level selection. It could be argued, for example, that altruistic behaviors are not really altruistic because they are promoted by rewards and punishments, making cooperation selectively advantageous within the group. However, the act of punishing is itself an altruist behavior, because it implies a risk and a cost of time and energy for the enforcer, while the whole group shares the benefits of the resulting order, so it could only have evolved through between-group selection. In the same way, if males in a group hunt and share food to obtain social rewards, we must explain the evolution of the social rewards, which itself is a multilevel selection problem (Wilson, 1998). In summary, whatever social mechanism we can attribute to promote altruist behavior must have had an initial fitness cost without a relative fitness benefit, so it could never have evolved through individual selection.

In biology, it is a well-stated fact that genes have evolved a regulatory system to largely eliminate intragenomic conflict, by actively suppressing “selfish” dominant genes. This genetic policing is, at its core, a form of social control. The underlying mechanics do not change when we shift up one level in the biological hierarchy. If genes suppressing other genes to help the body is considered “individual selection,” then individuals suppressing others to help the group is “group selection” (Wilson, 1997).

2.3. Mathematical models

As a first example, Fletcher and Zwick (2007) used a simple N-player prisoner’s dilemma to model the cooperator frequencies of two isolated groups in a computer simulation. As it was predicted, the frequency of cooperators increased whenever Hamilton’s rule was fulfilled, and decreased otherwise, for all parameter values. Since groups were isolated, defectors eventually took over every single group, making the variance ratio (r) plummet.

In a successive experiment, they periodically shuffled the population into new, random

groups. As it was found, applying the shuffle at the right time (around 20 generations), restored the variance ratio, creating a “ratchet effect” that pushed global cooperation up to 100% (Fletcher & Zwick, 2007).

Building on similar fitness functions, van Wijngaarden (2025) introduces a model that applies an elegant logic to keep both the overall number of groups and their individual sizes strictly constant. By pairing a weighted reproduction lottery with a random death event, the entire system can be visualized as a stochastic “ball-and-urn” process. Crucially, maintaining these constant sizes allows the multilevel selection dynamics to be analyzed purely through mathematics rather than simulations; as the population scales up, the ball-and-urn mechanics translate into partial differential equations that precisely calculate the thresholds where group-level selection dominates (van Wijngaarden, 2025).

In a later publication, Tarnita et al. (2009), among other proposed variations, develop the concept of ‘groups’ through overlapping, mutable sets. Sets represent shared traits, clubs or physical niches, and an individual can belong to more than one at a time. Offspring inherits their parent’s traits, with a probability of adopting random ones (set mutation). The authors’ major finding was that there is an optimum set mutation rate that strictly maximizes assortment (σ), allowing for cooperators to thrive and colonize other sets (Tarnita et al., 2009).

In their work, Tekwa et al. (2015) show that the idea of groups is entirely discarded to give way to a more realistic concept of spatial structure, where interaction probability is determined by the distance between two individuals. The authors define a metric to track cooperator assortment: the “clustering coefficient”. This metric is defined as the mean probability of a cooperator to interact with another cooperator compared to random chance, and is found to be determinant for the selection of the cooperative trait.

Finally, Szolnoki and Perc (2009) use random networks in a model where the network itself co-evolves with the agents. Agents do not spawn or die, instead they adopt the strategy of a successful neighbor. Links are deleted when an agent changes strategy, and new random links are added every τ steps. Their major finding was identifying an optimum rate for link addition ($\tau \approx 70$) that maximizes cooperator success. If τ is perfectly tuned, there is the right amount of interconnectedness with defectors to avoid being overrun by defectors, but enough to allow them to successfully invade defector groups.

3. THE MODEL

3.1. Model Definition

The proposed model synthesizes key elements from three of these frameworks with the goal of capturing the most essential form of biological mechanics across a variety of species.

First, we adopted the fitness and reproduction functions from Fletcher and Zwick (2007), allowing us to calculate individual fitness in a single step rather than simulating continuous agent-to-agent games. By avoiding the need for every agent to individually play the Prisoner’s Dilemma with every neighbor, we vastly reduce the computational load (Fletcher & Zwick, 2007).

Second, we incorporate Wiskunde’s (van Wijngaarden, 2025) use of a weighted reproduction lottery alongside a strictly constant population size. Preventing the population from scaling infinitely is crucial for keeping the model computationally efficient, while the mechanic preserves the exact same core dynamics of fitness and reproduction.

Finally, we utilize Tarnita et al. (2009)’s concept of overlapping sets over other forms of spatial structuring, as it provided an interesting and realistic idea of group belonging, with mechanics very similar to a network, while proving exceptionally simple to work with. Furthermore, sets are arguably the most generalizable topology in biology, representing simply an interaction dimension. As noted in Tarnita et al. (2009), a “set” could be a shared chemical receptor in microbial populations, a shared foraging territory in animal species, or even a specific flowering week for plant-insect interactions.

Below is a full explanation of the model, starting with a description of the used parameters:

Baseline fitness (w_0): The fitness floor value. It reduces the effect of the fitness payoff on reproductive success.

Cost (c): The absolute fitness penalty deducted from a cooperator to generate the benefit pool.

Benefit (b): The total value of the resource generated by a cooperator before distribution.

Total population (N): The absolute number of individuals in the population.

Total sets (M): The global pool of unique sets available in the environment.

Mean sets (K): The average number of sets assigned to an individual.

Set standard deviation (std): The standard deviation of set distribution among individuals.

Trait mutation (u): The probability of biological mutation, where an offspring randomly flips its base strategy (C to D, or D to C).

Set mutation (v): The probability of set mutation, where an offspring alters its inherited sets.

Directed exploration (x): A conditional probability triggering when v occurs. It dictates the chance an offspring selectively copies a set from a local neighbor instead of jumping to a random set.

Synergy (s): The scaling factor governing resource dilution. At its maximum value of 1, all set members receive the full benefit created by a cooperator.

As suggested above, the model consists of a space of overlapping groups or ‘sets’ containing individuals as members. An individual can be a member of multiple sets at the same time. This number is chosen for each individual at startup, and follows a normal distribution around K . Offspring will inherit their parent’s number of sets.

N individuals will spawn at startup. Half of them will possess the strategy “Cooperator”, and the other half will be “Defector”. Individuals will interact with other individuals they share sets with (neighbors) to collect a fitness payoff. If more than one set is shared, they interact that many times.

The payoff is defined by the following equation:

$$\text{payoff} = \frac{b \cdot k}{n^{1-s}} \quad (3.1)$$

$$(3.2)$$

Here, k represents the number of cooperators in the focal individual’s sets, and n is the total number of neighbors. The parameter s controls how much a cooperator’s benefit is diluted among the recipients, representing a range of expendable and non-expendable resources.

At each step, a random individual will die, and one of their neighbors will be chosen through a weighted lottery to reproduce. The ticket weight is defined through the reproduction equation:

$$\text{fitness} = w_0 + \text{payoff} \quad (3.3)$$

Here, w_0 is a baseline that can be used to control the weight of the payoff on reproduction. Offspring will inherit the parent’s strategy with a probability u of changing strategy, and it will inherit the parent’s sets, each set having a probability v of changing to a different set. Sets change randomly to a non-parent set, with a probability x of being chosen among the parent’s neighbors’ sets instead.

Set mutation mechanics point to the idea of groups trickling their influence over others in a controlled way. Since new connections are created by offspring, groups increase interconnectedness as they grow in size. The size and population density of the set network,

as well as the parameters that define group growth and expansion, will provide a greater advantage to cooperators or defectors, depending on the values of the parameters. This work is an initial attempt to unravel the logic of this difference. Because the challenge vastly exceeds the limited format, the task will be left as an avenue for future research in mathematical social network analysis.

Regarding the ‘synergy’ parameter: This novel parameter was added to represent various types of “public goods”. A synergy value of 0 represents a pure, expendable resource. A benefit of 10 divided by 10 people gives 1 to each. A value of 1 represents an infinitely reusable public good. A benefit of 10 gives 10 to everyone, regardless of the number of recipients. Finally, values in between represent a synergistic effect with diminishing returns. For example, with an s value of 0.5, a benefit of 10 among 10 people gives ≈ 3.16 . Among 20 people, it gives ≈ 2.23 . The dilution happens, but the penalty for adding more recipients softens.

Finally, conditional strategies were not considered in this model. While it seems logical that individuals would apply rules like “I will only cooperate if there are n other cooperators in my environment”, Wilson (1998) demonstrates that they have a homogenizing effect across the whole population, creating “below-random phenotypic variation among groups”. This means that conditional strategies can rarely be found in group-selection scenarios.

On the other hand, as it was already suggested, forms of social control, like punishments, rewards or means of cheater detection, are arguably the most powerful mechanisms in nature because they force conformity, suppressing within-group variation while amplifying between-group variation (Boehm, 1997, as cited in Wilson (1998)). However, these mechanics were left out due to their immense complexity. When social incentives are used, it can lead to “an infinite number of internally stable social conventions... ranging from highly adaptive to highly maladaptive at the group level” (Wilson, 1998). As fruitful as the enterprise might be, it must be left for future attempts. This model attempts to prove that cooperative traits can evolve in a population exclusively through simple mechanics of spatial assortment, as it is found in simpler lifeforms like bacteria, algae and, arguably, even pre-biotic molecule structures (Tarnita et al., 2009).

3.2. Methodology

The agent-based model was developed in NetLogo to simulate complex evolutionary dynamics. Parameter sweeping was conducted using BehaviorSpace, NetLogo’s integrated experimental tool, to systematically run the simulation across a number of parameter combinations across the parameter space. Given the high computational demands of the model, simulation executions were offloaded to a cloud virtual private server.

Despite the advantages of cloud computing, holding specific parameters constant was needed to avoid massive computational loads. Population size (N), individual cost (c),

and baseline fitness (w_0) are likely to exert negligible effects on the mechanics of this model, as suggested by preliminary analysis and established theoretical frameworks (van Wijngaarden, 2025), although measuring their potential effects on the model remains a valid subject for future research. Two other parameters, trait mutation (u) and set standard deviation (std), were confirmed to have relevant effects on the outcome, but were excluded from the analysis due to the aforementioned constraints. As a result, the following values were fixed in all simulations: $N = 200$, $c = 10$, $w_0 = 10$, $u = 0$, and $std = 2$. Since s was found to have an immense influence on the outcome, three separate simulations were run for three of its values ($s = 0$, $s = 0.5$, and $s = 1$), choosing the remaining parameters accordingly to scan the parameter space areas where cooperation is contested, and avoiding scanning the vast areas where cooperation always fails or always succeeds. Additionally, one final experiment was run choosing a finer selection of s and b values to measure their specific interaction, while keeping the remaining parameters constant at $x = 0$, $K = 12$, $M = 150$, and $v \in \{0, 0.25, 0.5\}$.

Simulation outputs in csv format were loaded into RStudio and aggregated into summary metrics prior to statistical analysis. The primary dependent variable, cooperation success (`coop_success`), was quantified as the mean frequency of cooperators over the final 2000 steps of each run, bounding the outcome between 0 (complete defection) and 1 (altruism fixation). Independent variables were formalized as ratios to evaluate proportional relationships, specifically the network density ratio (K/M) and the cost-benefit ratio (b/c).

To capture the interactions of the simulated parameters, the aggregated dataset was used to train a Random Forest regression model. Random Forest is an ensemble machine learning algorithm that constructs a multitude of decision trees, recursively locating the variable and value that best divides the dataset to maximize the mean difference in outcome between the two subsets. This method is effective for non-linear interactions like the ones expected in the data. Furthermore, its process and output are highly interpretable.

The relative importance of the various parameters was calculated through permutation variable importance. This technique measures the drop in a model's predictive accuracy when the data for a specific feature is randomly shuffled. By quantifying this accuracy loss, the algorithm objectively ranked the parameters that most strongly dictated cooperation success or failure.

To map the boundaries of cooperation success across the parameter space, the data was processed using Locally Estimated Scatterplot Smoothing (LOESS). LOESS is a non-parametric regression method that applies sliding, localized polynomial fits to generate smooth curves or continuous surfaces from noisy data.

One-dimensional LOESS models were applied to track shifting optimal points, allowing the extraction of the exact mathematical peaks—such as the specific K/M density required to maximize cooperation under various mutation rates.

Two-dimensional LOESS models were employed to construct three-dimensional per-

formance landscapes. By extracting specific contour lines (isoclines) from these surfaces, the analysis mapped exact evolutionary frontiers. Measuring the slope of an isocline at a strict success threshold (e.g., 50% or 80% `coop_success`) provided the precise exchange rate between two variables required to prevent the collapse of the cooperative population.

3.3. Analysis

This section presents the results of the computational simulations, evaluating the parameter thresholds required for cooperation to evolve. To isolate the effects of resource dynamics, the analysis is segmented across three distinct synergy environments: pure expendable resources ($s = 0$), diminishing returns ($s = 0.5$), and infinite public goods ($s = 1$).

The analysis proceeds in three stages:

1. **Predictive Hierarchy:** Random Forest variable importance and decision trees evaluate and rank the structural parameters that most strongly dictate cooperation success.
2. **Synergy Trade-offs:** Two-dimensional LOESS models map the evolutionary boundaries between the cost-benefit ratio and synergy under varying rates of cultural mutation.
3. **Density Frontiers:** Continuous LOESS surfaces define the precise exchange rates required between network density (K/M) and the cost-benefit ratio to sustain a cooperative population.

3.4. Variable Importance

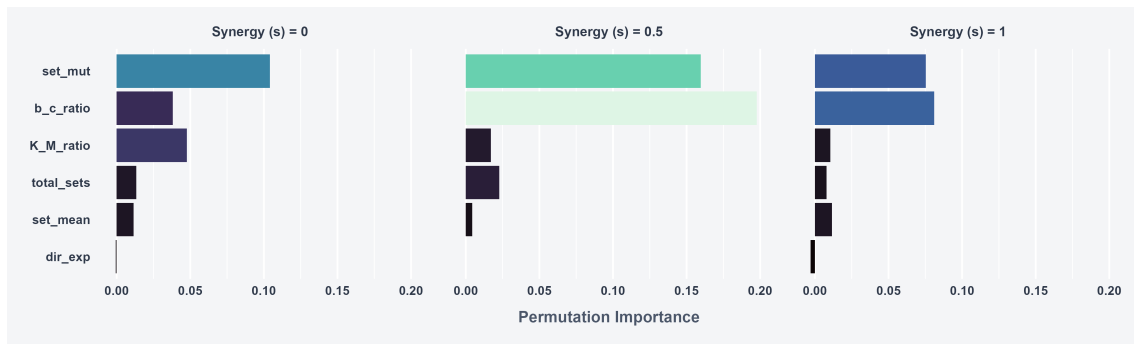


Fig. 3.1. *Random Forest Variable Importance*

The Random Forest permutation importance reveals how the primary drivers of cooperation shift across synergy levels.

In environments with pure expendable resources ($s = 0$), set mutation (v) is the strongest predictor of cooperation success. Network density (K/M) and the cost-benefit ratio (b/c) follow as secondary factors. As we will see, the rate of random group exploration is very beneficial for defectors at this level of synergy.

When synergy is introduced partially ($s = 0.5$) or totally ($s = 1$), the b/c ratio and set mutation dominate the predictive hierarchy. This result highlights the power of these variables: high cost-benefit ratios reliably drive cooperation, while high mutation rates consistently destroy it.

However, this global dominance does not render network density K/M irrelevant, while this variable alone cannot overpower extreme mutation or abundant resources, its behavior can still be studied under moderate conditions, requiring the closer boundary analysis presented in the following sections.

Regarding directed exploration (x), it registered no effect across any of the measures of this analysis. Consequently, x is excluded from further analysis, leaving its dynamics for future investigation.

3.4.1. Decision Tree Analysis

To understand how these variables interact to create distinct evolutionary pathways, a single decision tree was extracted for each synergy level. These trees illustrate the specific numerical thresholds that most dramatically partition the parameter space into cooperative or defective outcomes.

Pure Expendable Resources ($s = 0$)

The decision tree in Figure 3.2 highlights the extreme fragility of cooperation in the absence of synergistic returns. The root node splits the data at a severely low set mutation threshold ($v = 0.075$). If mutation exceeds this minimal rate, mean cooperation success collapses to 0.052 or 5.2%.

Tracing the high-mutation branch reveals how extreme network sparsity ($K/M < 0.054$) can mitigate the effects of set mutation, bringing cooperation success to 0.24. Under these conditions, highly cooperative clusters are able to stay isolated from the active search of defectors.

Conversely, the right branch maps the strict requirements for cooperative dominance. Suppressing set mutation ($v < 0.075$) establishes a baseline success of 0.42. To improve upon this, the network must still maintain a relatively low density ($K/M < 0.11$), which elevates success to 0.74. even within this restricted, low-mutation, sparse-network topology a moderate cost-benefit ratio ($b/c \geq 2.8$) is still critical for the cooperative trait to succeed, as values under 2.8 make cooperation success go down to 0.33.

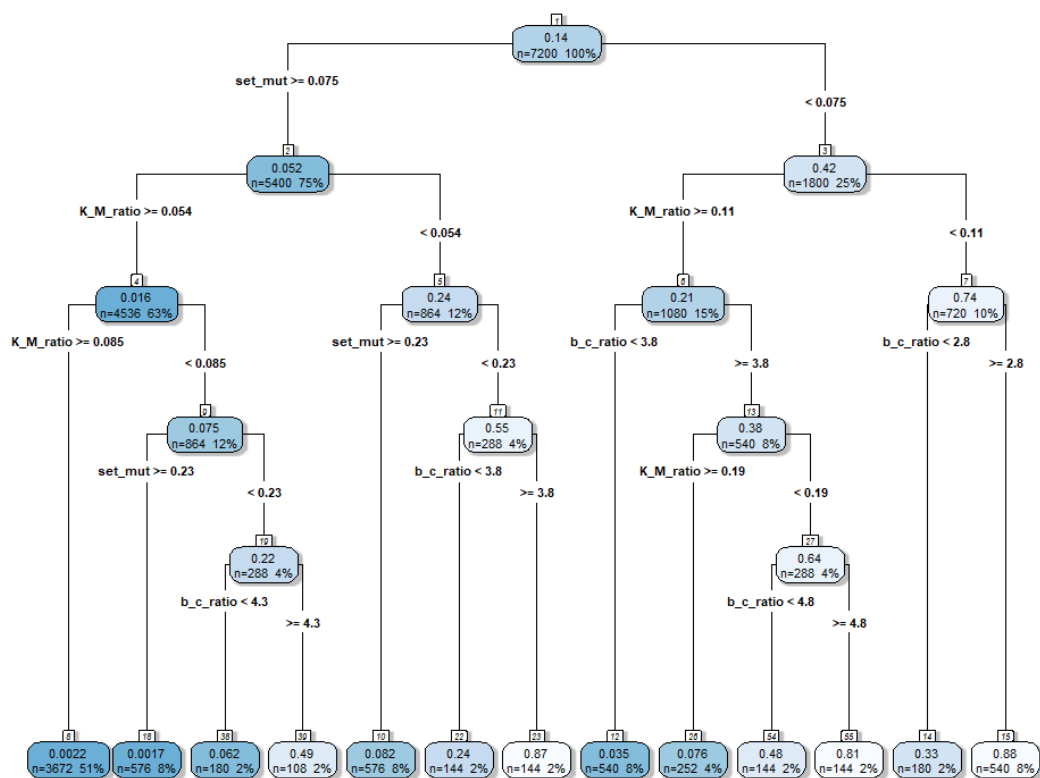


Fig. 3.2. Decision Tree. Synergy = 0

Partial Synergy ($s = 0.5$)

In Figure 3.3 we see how introducing partial synergy alters the primary driver of cooperation. The root node partitions the data based on the cost-benefit ratio at a threshold of $b/c = 0.3$. This demonstrates how synergy drastically reduces the cost-benefit ratio required for altruism to spread.

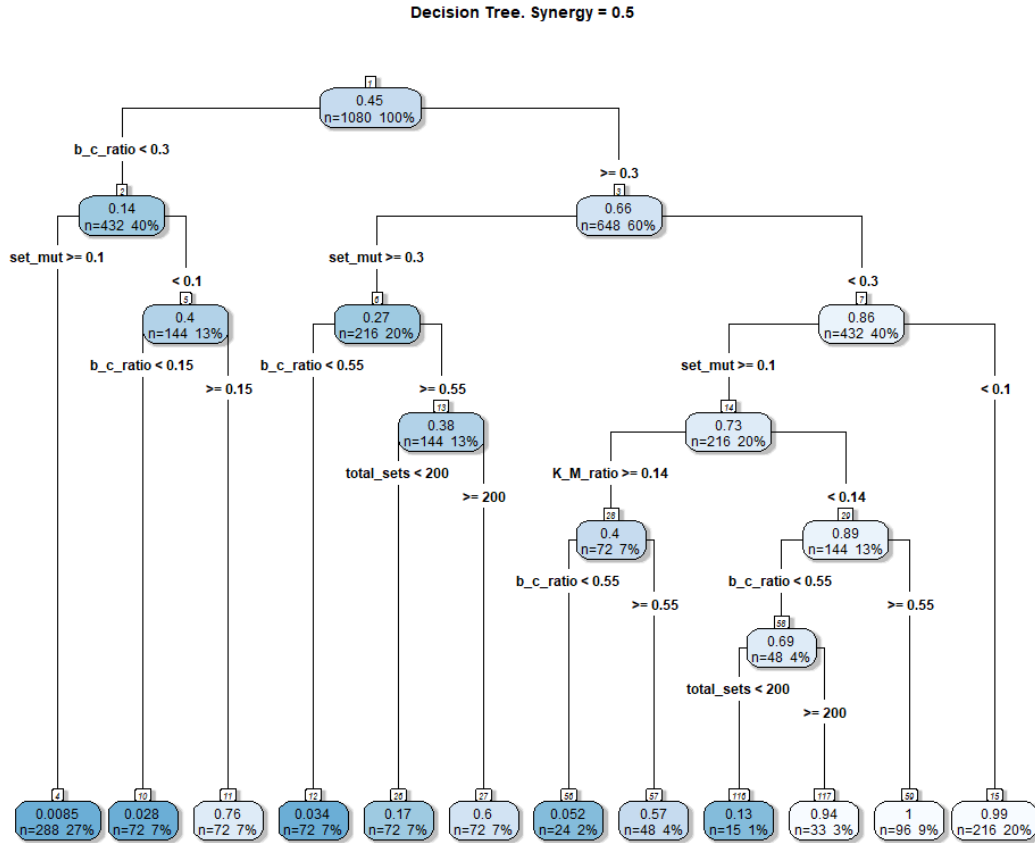


Fig. 3.3. *Decision Tree. Synergy = 0.5*

When resources are relatively scarce ($b/c < 0.3$), average success falls to 0.14. Yet, cooperation can recover if set mutation is highly restricted ($v < 0.1$). Under these stable conditions, a minor increase in resource efficiency ($b/c \geq 0.16$) drives success back up to 0.76.

When $b/c \geq 0.3$, baseline success improves to 0.66, but remains highly vulnerable to set mutation, with values of ($v \geq 0.3$) degrading the cooperative population down to 0.27. Restricting mutation ($v < 0.3$) secures a robust success rate of 0.86, and minimizing it further ($v < 0.1$) achieves near-fixation at 0.99.

For moderate mutation rates ($0.1 \leq v < 0.3$), network density shows a critical threshold at ($K/M < 0.14$), with runs under threshold showing more than double cooperator success. However, it is shown that extreme spatial isolation is no longer strictly necessary

to protect cooperators when resources possess synergistic properties. Additionally, the absolute number of sets emerges as a secondary splitting variable, indicating a moderate positive effect on cooperation in specific cases that warrants future targeted study.

Total Synergy ($s = 1$)

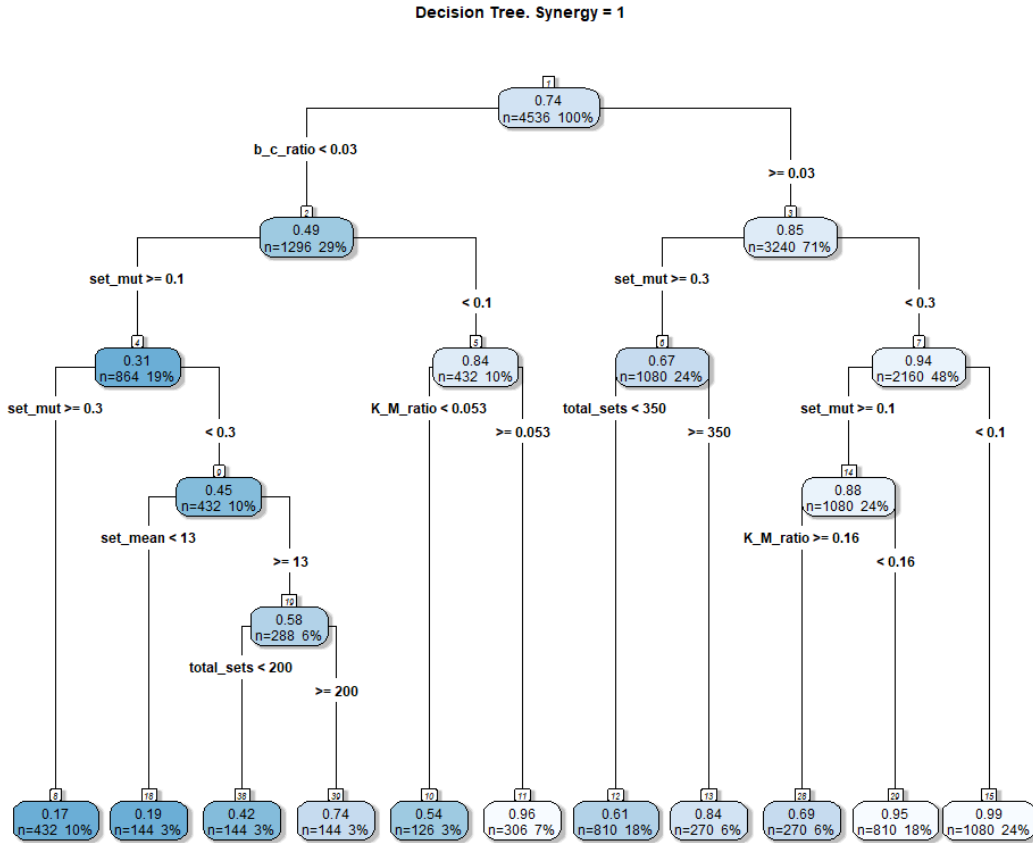


Fig. 3.4. *Decision Tree. Synergy = 1*

Total synergy maximizes resource value in Figure 3.4, causing the root node to split at a very low cost-benefit ratio ($b/c = 0.03$).

When resources meet this minimal threshold ($b/c \geq 0.03$), average success reaches 0.85 and becomes highly resilient. Even with high set mutation and high network density, success only degrades to 0.61. Under more stable conditions ($v < 0.3$), cooperation thrives at 0.94. For moderate mutation rates ($0.1 \leq v < 0.3$), maintaining a network density of $K/M < 0.16$ secures a success rate of 0.95.

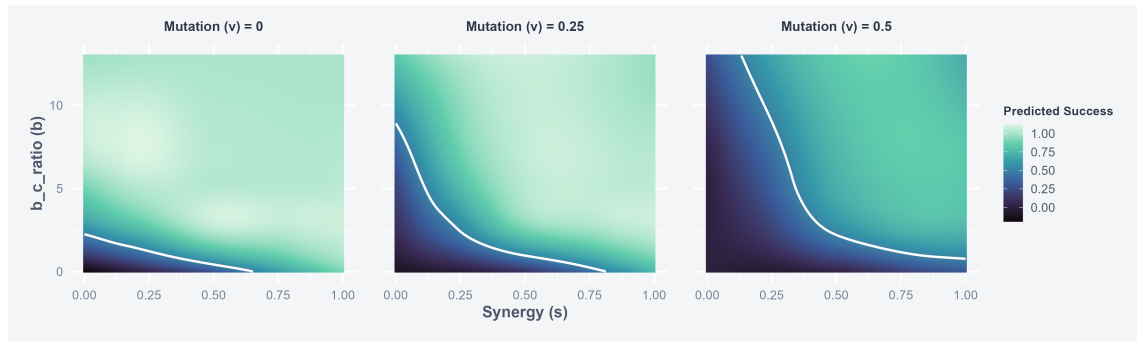
When resources are constrained ($b/c < 0.03$), average success drops to 0.49, and set mutation becomes the critical determinant. If mutation is elevated ($v \geq 0.1$), success falls to 0.31 and is difficult to recover. However, if mutation is strictly limited ($v < 0.1$), success rebounds to 0.84. Within this restricted sub-branch, network density acts as the

final splitting threshold: a highly sparse network ($K/M < 0.053$) pushes success to 0.96, compared to 0.54 in denser configurations.

Ultimately, this demonstrates that even under extreme resource scarcity, cooperation can achieve fixation if the population maintains strict spatial assortment.

3.4.2. 2D LOESS Plots

b/c Ratio vs Synergy



Note: The white line represents 50% success.

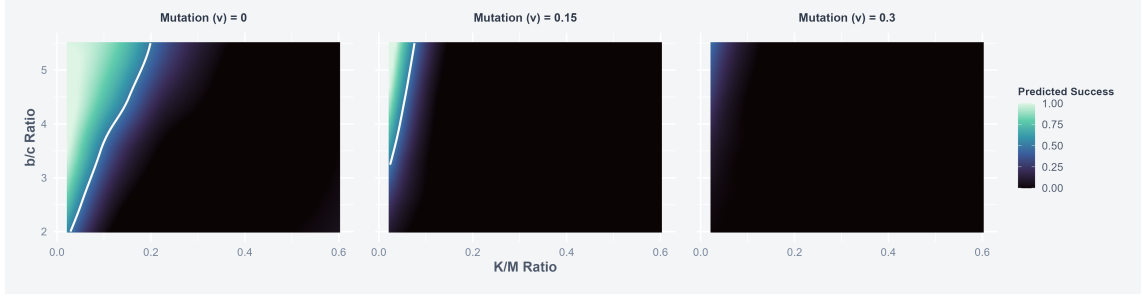
Fig. 3.5. *b/c* Ratio vs Synergy

The two-dimensional LOESS surfaces map the interaction between synergy and the cost-benefit ratio across three mutation rates, the white line represents 0.5 or 50% success rate.

In absence of set mutation ($v = 0$), cooperation dominates the parameter space. The isocline reveals an inverse relationship: as synergy increases, the required b/c ratio drops slowly, reaching a point where cooperators succeed even with minimal cost-benefit ratios.

Introducing moderate mutation ($v = 0.25$) low-synergy environments are disproportionately affected, forcing a steep increase in the b/c ratio needed to prevent cooperative collapse. However, high-synergy environments demonstrate robust resilience, maintaining the survival threshold nearly unaffected.

Under high set mutation rates ($v = 0.5$), the dynamic aggravates. While high synergy environments show resilience, the whole 2d space is affected and there are no high success points. Peak cooperation is capped at approximately 0.75, even at high synergy and cost-benefit ratios.



Note: The white line represents 50% success.

Fig. 3.6. b/c Ratio vs K/M Ratio. Synergy = 0

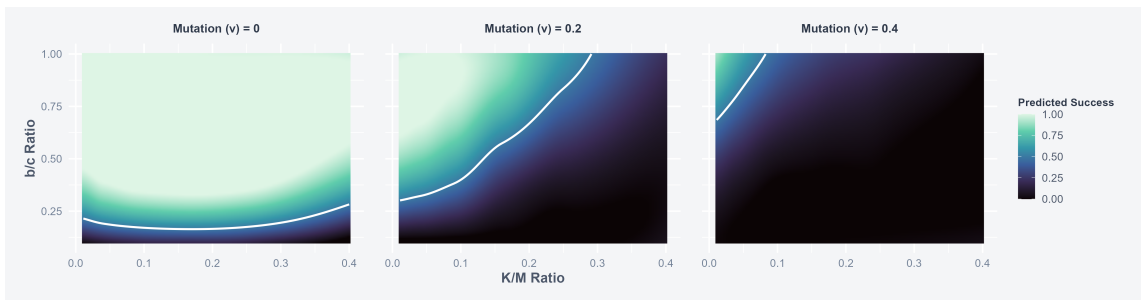
3.4.3. b/c Ratio vs K/M Ratio

Pure Expendable Resources ($s = 0$)

The plots for $s = 0$ map a steep trade-off between network density and resource efficiency. In the absence of set mutation ($v = 0$), cooperation can succeed at very low b/c ratios in sufficiently sparse networks. As network density (K/M) increases, the required b/c ratio rises sharply to maintain success.

Introducing moderate mutation ($v = 0.15$) steepens the relation, meaning sparsity is even more determinant, and the viable cooperative zone is compressed. Under high mutation ($v = 0.3$), cooperation collapses entirely within the evaluated parameter space, but success might be possible for other values. Capturing potential cooperative success at this set mutation level would require future experiments testing more extreme parameter boundaries.

Partial Synergy ($s = 0.5$)



Note: The white line represents 50% success.

Fig. 3.7. b/c Ratio vs K/M Ratio. Synergy = 0.5

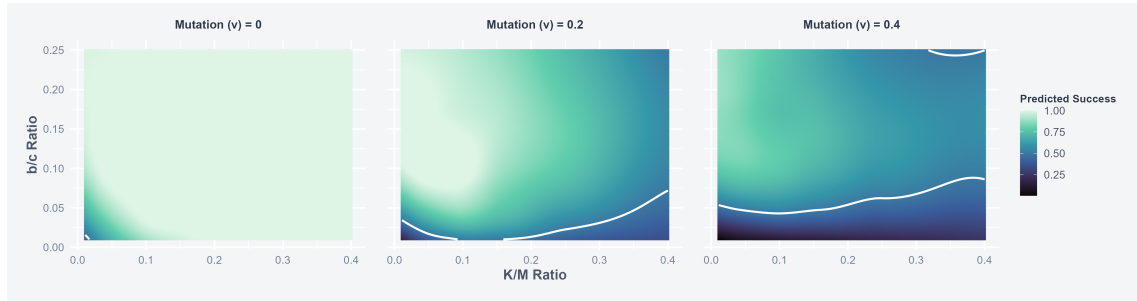
At this level of synergy we find a crucial difference. In the absence of set mutation ($v = 0$), the trade-off between network density and resource efficiency largely disappears. A b/c ratio of approximately 0.25 maintains cooperative success across the entire evaluated

space, with only minor increases required at the extreme high and low ends of network density.

Introducing moderate mutation ($\nu = 0.2$) restores the familiar structural dynamic: higher network interconnectedness requires a higher b/c ratio to prevent cooperative collapse, mapping a clear diagonal frontier.

Under high mutation ($\nu = 0.4$), this relationship steepens, compressing the viable cooperative zone into sparse, high-benefit environments. As observed in the previous graph, capturing reliable cooperation under such high cultural volatility requires future research exploring more extreme parameter boundaries.

Total Synergy ($s = 1$)



Note: The white line represents 50% success.

Fig. 3.8. b/c Ratio vs K/M Ratio. Synergy = 1

At this level of synergy, environments with no set mutation appear heavily dominated by cooperation. The 50% success frontier falls mostly out of frame, as minimal resource efficiency is sufficient to achieve fixation across nearly all network densities. This is a scale problem, and the plot suggest that adjusting the scale would show a roughly diagonal inverse linear relation between the two variables. As opposed to the other cases, extreme isolation proves detrimental at very low cost-benefit ratios, likely because it prevents the effective distribution of the non-depleting resource.

Under moderate mutation ($\nu = 0.2$), cooperative success is globally hindered, but a high-success area at $K/M < 0.1$ persists. At high mutation ($\nu = 0.4$), cooperation suffers, but the same area shows resilience at around 80% success. This level of synergy shows extreme resilience across most of the evaluated parameter space.

Conclusions

This thesis recapitulates how Multilevel Selection (MLS) provides a robust mathematical framework for understanding the evolution of altruism, driven by the tension between within-group selfishness and between-group variation. By utilizing a computational agent-based model based on overlapping sets, this research contributes to mapping

the specific structural and environmental variables required for group-level selection to dominate, as well as the specific relations between them.

The analysis reveals that synergy, cost-benefit ratio and set mutation rate are the most powerful global predictors of cooperative success. For non-synergistic traits, cooperation is highly determined by spatial isolation and controlled spreading. For all parameter values, the capacity of individuals to infiltrate random sets has a negative effect on cooperation. Higher synergy values show much higher resilience under a range of conditions, removing the need from spatial isolation and allow cooperative traits to thrive even within highly interconnected networks. Ultimately, this model confirms that altruism can reliably evolve and achieve fixation exclusively through simple mechanics of spatial assortment, provided the right values for cost, benefit and network sparsity are present.

BIBLIOGRAPHY

- Borrello, M. E. (2005). The rise, fall and resurrection of group selection. *Endeavour*, 29(1), 43–47. <https://doi.org/10.1016/j.endeavour.2004.11.003>
- Dawkins, R. (1976). *The selfish gene*. Oxford University Press.
- Fletcher, J. A., & Zwick, M. (2007). The evolution of altruism: Game theory in multi-level selection and inclusive fitness. *Journal of Theoretical Biology*, 245(1), 26–36. <https://doi.org/10.1016/j.jtbi.2006.09.030>
- Gavrilets, S. (2015). Collective action problem in heterogeneous groups. *Philosophical Transactions of the Royal Society B*, 370, 20150016. <https://doi.org/10.1098/rstb.2015.0016>
- Greene, C. A., McPeck, S. J., Mitchem, L., Clark, A. B., Formica, V. A., Philson, C. S., & Costello, R. A. (2025). Multilevel selection in the margins: A review of its representation in undergraduate biology textbooks. *Ecology and Evolution*, 15, e72493.
- Hamilton, W. D. (1964). The genetical evolution of social behaviour. i and ii. *Journal of Theoretical Biology*, 7(1), 1–52. [https://doi.org/10.1016/0022-5193\(64\)90038-4](https://doi.org/10.1016/0022-5193(64)90038-4)
- Marín, C., Clark, A. B., Philson, C. S., Eldakar, O. T., & Wade, M. J. (2026). Abundant empirical evidence of multilevel selection revealed by a bibliometric review. *Frontiers in Ecology and Evolution*, 14, 1752597. <https://doi.org/10.3389/fevo.2026.1752597>
- Szolnoki, A., & Perc, M. (2009). Emergence of multilevel selection in the prisoner's dilemma game on coevolving random networks. *arXiv*.
- Tarnita, C. E., Ohtsuki, H., Antal, T., Fu, F., & Nowak, M. A. (2009). Strategy selection in structured populations. *Journal of Theoretical Biology*, 259(3), 570–581. <https://doi.org/10.1016/j.jtbi.2009.03.035>
- Tekwa, E. W., Gonzalez, A., & Loreau, M. (2015). Local densities connect spatial ecology to game, multilevel selection and inclusive fitness theories of cooperation. *Journal of Theoretical Biology*, 380, 414–425. <https://doi.org/10.1016/j.jtbi.2015.06.016>
- van Wijngaarden, F. (2025). *Modeling selection at the individual and group level* [Bachelor's thesis]. Universiteit Utrecht.
- Wilson, D. S. (1997). Altruism and organism: Disentangling the themes of multilevel selection theory. *The American Naturalist*, 150(S1), S122–S134. <https://doi.org/10.1086/286053>
- Wilson, D. S. (1998). Hunting, sharing, and multilevel selection: The tolerated-theft model revisited. *Current Anthropology*, 39(1), 73–97. <https://doi.org/10.1086/204699>
- Wilson, D. S., Madhavan, G., Gelfand, M. J., Hayes, S. C., Atkins, P. W. B., & Colwell, R. R. (2023). Multilevel cultural evolution: From new theory to practical applications. *Proceedings of the National Academy of Sciences*, 120(16), e2218222120. <https://doi.org/10.1073/pnas.2218222120>

- Wilson, D. S., & Wilson, E. O. (2007). Rethinking the theoretical foundation of sociobiology. *The Quarterly Review of Biology*, 82(4), 327–348. <https://doi.org/10.1086/522809>
- Wynne-Edwards, V. C. (1962). *Animal dispersion in relation to social behaviour*. Oliver & Boyd.

ANNEX I

All used data and code can be found in the following GitHub repository:

<https://github.com/IggyWake/ebb-and-flow/>

ANNEX II

DECLARATION OF USE OF GENERATIVE ARTIFICIAL INTELLIGENCE (AI) IN MASTER FINAL PROJECT (TFM)

I have used Generative AI in my TFM

Check all that apply:

YES	☐
-----	--------------------------

If you have ticked YES, please complete the following 3 parts of this document:

Part 1: Reflection on legal, ethical and responsible behavior

Please be aware that the use of AI carries some risks and may lead to a series of serious academic consequences: the TFM will not be evaluated by the University if the use of AI involves the use of confidential data, copyrighted materials, or personal data, used without complying with the conditions required in each case (authorization of the interested parties, authorization of the owners, following the instructions of the University).

Question	
1. In my interactions with AI tools, I have provided confidential data, always with the appropriate authorization of the data subjects. Confidentiality encompasses any information that a person or organization wishes to protect for legal, commercial, privacy, or strategic reasons (such as patents or trade secrets).	
YES, I have used this data with permission	NO, I have not used confidential data X
2. In my interaction with Generative AI tools, I have submitted copyrighted materials with the permission of those concerned.	
YES, I have used these materials with permission from the copyright holders, or without permission because they fall within one of the exceptions or limitations permitted by law: • Public domain work • Licensed work (Creative Commons licenses) • Use of excerpts for research purposes (fair use)	NO, I have not used copyrighted materials X

3. In my interaction with Generative AI tools, I have submitted personal data with the consent of the data subjects.	
YES, I have used this data with the authorization of the interested parties and in accordance with the instructions contained in the guide approved by the University .	NO, I have not used personal data X
4. My use of the Generative AI tool has respected its terms of use , as well as the essential ethical principles, not being maliciously oriented to obtain an inappropriate result for the work presented, that is to say, one that produces an impression or knowledge contrary to the reality of the results obtained, that supplants my own work or that could harm people.	
YES	NO X

Part 2: Declaration of technical use

Use the following model statement as many times as necessary, in order to reflect all types of iteration you have had with Generative AI tools. Include one example for each type of use where indicated: *[Add an example]*.

I declare that I have made use of the Generative AI systems Gemini and Notebook LM for:

Documentation and drafting:

- *Supporting reflection in relation to the development of the work: iterative process of analysing alternatives and approaches using AI*

I have discussed with Gemini about the nature of the problem and possible approaches for the investigation

- *Revision or rewriting of previously drafted paragraphs*

For each section, I provided Gemini with a draft with all the ideas and citations, and it fixed the writing style. Afterwards, I applied several corrections, deleting and adding words and sentences

- *Search for information or answers to specific questions*

I used Notebook LM for bibliography search and consultation

Develop specific content

Generative AI has been used as a support tool for the development of the specific content of the TFM, including:

- *Assistance in the development of lines of code (programming)*

I have used Gemini to produce code for both Netlogo and R, always asking for precise explanations of each line and personally reviewing and testing the code thoroughly.

- *Generation of diagrams, images, audios or videos*

I have used Notebook LM to generate audios with the content of some reference papers to get familiar with it

- *Optimisation processes*

I have used Gemini to correct errors in code that was not running properly

- *Data processing: collection, analysis, cross-checking of data...*

I have used Gemini to write code for cleaning, summarizing and analyzing data

- *Inspiration of ideas in the creative process*

I asked Gemini to suggest appropriate methods for the desired inquiry about the data.

Part 3: Reflection on utility

Please provide a personal assessment (free format) of the strengths and weaknesses you have identified in the use of Generative AI tools in the development of your work. Mention if it has helped you in the learning process, or in the development or drawing conclusions from your work.

Mostly AI tools have helped me with technical issues regarding code, mathematics and software use, as well as fixing the writing style for the final product. The field where they have performed best is at solving and explaining complex mathematical problems. It worked out the mathematical proof needed to convert data that I had into metrics that I needed for the analysis, and after careful review, everything was correct. This is the field where it has helped me gain more knowledge and practice.
A negative point I would highlight is the use of excessive adjectives or 'bloat' language when writing in academic style, which looks poor and unnecessarily complicates the ideas.